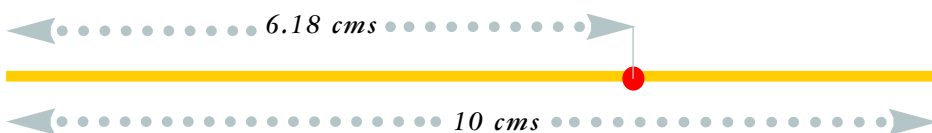




## WHAT IS PHI?

Draw a straight line 10 cm long, then make a small mark on it 6.18 cm along. You've divided the line into two sections. If you divide the length of the whole line by the length of the long section, you'll get the number 1.618. And if you divide the length of the long section by the length of the short section, you'll get the same ratio. This is the *golden ratio*, or phi, written  $\Phi$ .



### Phantastic phi

Phi has strange properties. Multiplying it by itself, for instance, is exactly the same as adding one. If you divide any number in the Fibonacci series by the one before, you'll get a ratio close to phi. This ratio gets closer to phi as you travel along the series, but it never quite gets there. In fact, it's impossible to write phi as a ratio of two numbers, so mathematicians call it "irrational". If you tried to write phi as a decimal, its decimal places would go on forever.

$$1 \div \Phi = \Phi - 1$$

$$\Phi \times \Phi = \Phi + 1$$

## FAQ

### What's magic about phi?

Ancient Greeks thought phi was magic because it kept cropping up in shapes they considered sacred. In a five-pointed star, for instance, the ratio between long and short lines is phi exactly.

### Why did artists use phi?

Leonardo da Vinci and other artists of medieval Europe were fascinated by maths. They thought shapes involving phi had the most visually pleasing proportions, so they often worked them into paintings.

### Building with phi

Ancient Greek architects are said to have used phi in buildings. Some people claim the Parthenon (below) in Athens is based on golden rectangles. What do you think?

